Par Bahrae

Quantitative Research · Data Analysis · Risk Analytics

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SUMMARY

Mathematics (Honours) graduate with statistics research experience who builds reproducible quantitative research on real market data: DCF valuation, portfolio risk, option-implied volatility and walk-forward volatility forecasting. Forecasting and portfolio models are tested out-of-sample against benchmarks and naive baselines, with results reported as found, including when the simple model wins. Brings rubric-based evaluation rigor from LLM evaluation work and statistical modelling in R and SAS.

SELECTED PROJECTS

Portfolio Risk & Performance Analyzer | Python, SQL, Streamlit, pytest, CI · minimum-variance optimization — Live app

- 2007–2026: min-variance cut max drawdown to −19.7% vs −55.2% for SPY, at a lower Sharpe (0.35 vs 0.55), reported as found.

Portfolio Risk Lab | Python, React · mean-variance, risk parity, Black-Litterman, VaR/CVaR — Live demo

- Risk parity cut max drawdown to −12.72% vs −23.93% for SPY (21 ETFs, walk-forward, net of costs); none beat SPY's 0.92 Sharpe.

Volatility Forecasting ML | HAR-RV, GARCH(1,1), gradient boosting, LSTM (PyTorch), Diebold-Mariano — Live demo

- LSTM cut mean QLIKE 9.0% vs HAR (0.2968 vs 0.3262) over 3,473 out-of-sample days on 10 ETFs, significant on 8 of 10; no MSE edge.

Options Volatility Surface | Black-Scholes-Merton, Greeks, implied-vol solver, SVI, arbitrage checks — Live demo

- Fitted SVI to 1,077 SPY quotes across 11 expiries: 0.22–0.67 vol-point RMSE per slice, butterfly-arbitrage-free on 11 of 11 slices.

DCF Valuation Model | Python, TypeScript/React · FCFF DCF, WACC/CAPM, Monte Carlo, reverse DCF — Live demo

- Valued 6 US large caps from SEC XBRL filings; TypeScript engine matches Python within 0.01% and runs a 10,000-draw Monte Carlo.

UNH Long Equity Pitch | Two-stage DCF, reverse DCF, comps, Fama-French factors, walk-forward backtest — Live demo

- Base-case DCF value $526.64 vs a $367.08 price (+43.5%); the market price implies 6.0% FCF growth vs a 13.7% base case.

EXPERIENCE

Mercor — *Generalist Expert*, Remote — Aug 2026 – Present

- Evaluate and rank AI responses for factual accuracy, reasoning quality, relevance, clarity and instruction-following.
- Apply evaluation rubrics and annotation guidelines to flag hallucinations, safety risks and quality defects; give actionable feedback.

Invisible Technologies — *Master - Persian Language Specialist - AI Trainer*, Remote — Aug 2025 – Aug 2026

- Created and validated Persian datasets in Python and PostgreSQL, checking grammar, meaning, cultural relevance and consistency.
- Evaluated LLM outputs for accuracy, safety and instruction-following; annotated quality signals and failure cases, and reviewed dataset workflows for quality assurance.

Toronto Metropolitan University (TMU) — *Research Assistant, Department of Statistics*, Toronto, ON — May 2023 – Aug 2025

- Analyzed research datasets with R and SAS and built predictive models to support forecasting and research objectives.
- Prepared reports and presentations explaining statistical findings to faculty and other stakeholders.

Toronto Metropolitan University (TMU) — *Junior AI Engineer*, Toronto, ON — Sep 2022 – 2024

- Built and tested research tools with Python, C/C++ and SQL; debugged software and analyzed project data.
- Improved workflows through testing and documentation; collaborated with faculty and researchers to communicate findings to technical and non-technical audiences.

SKILLS

Languages: Python, SQL, R, SAS, TypeScript, JavaScript, C/C++

Quant/Stats methods: DCF and reverse DCF, WACC/CAPM, Monte Carlo, minimum-variance, mean-variance, risk parity, Black-Litterman, VaR/CVaR, Fama-French factor regression, Black-Scholes Greeks, SVI, HAR-RV, GARCH(1,1), LSTM, gradient boosting, Diebold-Mariano tests, walk-forward backtesting

Data/Tools: pandas, NumPy, scikit-learn, PyTorch, PostgreSQL, SQLite, Streamlit, React, pytest, GitHub Actions CI, Excel/Google Sheets, Jira, Unix/Linux, VMware, Visual Studio, ChatGPT

Evaluation & QA: Rubric-based scoring, side-by-side comparison, RLHF preference ranking, hallucination detection, dataset validation, error analysis, Persian linguistic review

EDUCATION

Toronto Metropolitan University (TMU) — Bachelor of Science, Mathematics and Its Applications (Honours) — Jan 2022 – Jan 2025

Minor in CS & Politics · GPA: A- · Thesis: Applications of Statistics and Probability

Leadership: Vice President, Department Student Association

Coursework: Data Analysis, Probability Theory, Linear Algebra, Database Systems, Statistics